How can India expand AYUSH education while ensuring credible training and effective quality regulation?

Capacity Gap • Clinical Exposure • Faculty Shortfalls • Regulatory Oversight • Incentive Misalignment

GS Paper II
Health
Government Policies

KNOWLEDGE POINTS

- AYUSH Education System: AYUSH medical education covers traditional and alternative systems such as Ayurveda and homoeopathy, requiring specialised colleges, teaching hospitals, qualified faculty, clinical exposure and appropriate regulatory oversight.
- Private-Sector Expansion: Private institutions increasingly drive AYUSH education growth, creating additional capacity but also raising concerns that financial incentives may encourage higher student intake without proportionate investment in faculty and infrastructure.
- Quality Capacity Gap: Rapidly increasing seats can strain educational quality when teaching staff, laboratories, hospital beds and clinical services do not expand proportionately, weakening students' practical exposure and institutional capacity.
- Clinical Training Quality: Medical education depends not merely on classroom teaching but meaningful clinical exposure; inadequate hospital facilities, patient attendance or bed occupancy can restrict students' opportunities to develop practical competence.
- Quality-Control Reform: Sustainable expansion requires stronger inspections, credible faculty verification, enforcement of admission limits and infrastructure standards, alongside regulatory mechanisms that align institutional incentives with educational quality.

QUICK FACTS

- In 2024, non-government institutions accounted for 86% of Ayurveda colleges and 85% of homoeopathy colleges, demonstrating private-sector dominance in AYUSH education.
- Between 2021 and 2024, permitted AYUSH seats increased by 43%, while total admission capacity rose by 25%, indicating rapid expansion of medical education.
- During 2021–2024, the Centre's AYURGYAN allocation for AYUSH education, training, research, innovation and capacity-building increased nearly sixfold alongside expanding educational capacity.
- A 2020 Journal of Ayurveda and Integrative Medicine article reported that many institutions had teaching-staff shortfalls exceeding 50% against the standards then applicable.

KEY TAKEAWAY

Expanding AYUSH education sustainably requires growth in seats to be matched by qualified faculty, clinical exposure, adequate infrastructure and rigorous regulatory oversight.

How can India protect recovering vulture populations from emerging threats posed by expanding power infrastructure?

Ecological Scavengers • Diclofenac Threat • Electrocution Risk • Bird-Safe Infrastructure • Preventive Planning

GS Paper III
Environment
Biodiversity
Conservation and Infrastructure

KNOWLEDGE POINTS

- Ecological Scavengers: Vultures rapidly consume animal carcasses, limiting their persistence in the environment and thereby helping regulate scavenger populations, disease transmission risks and broader ecosystem sanitation.
- Electrocution Mechanism: Vultures can be electrocuted when their large wingspans simultaneously bridge energised conductors or connect energised and grounded components, allowing dangerous electrical current to pass through them.
- Behavioural Vulnerability: Vultures frequently perch on elevated structures and congregate around predictable food sources, making electrical installations particularly hazardous when carcasses or food waste attract birds nearby.
- Infrastructure Expansion Risk: Expanding electricity infrastructure can create new wildlife hazards when transmission and distribution systems are designed without considering bird behaviour, body dimensions and ecologically sensitive locations.
- Bird-Safe Infrastructure: Separating feeding sites from electrical infrastructure, insulating conductors, increasing component spacing and installing safe perches can reduce opportunities for vultures and other birds to complete electrical circuits.

QUICK FACTS

- India's vulture population declined by 99.5% by 2007 from around four crore in the 1980s, severely affecting several major vulture species.
- Diclofenac, a veterinary painkiller that causes kidney damage in vultures feeding on treated cattle carcasses, was identified as the principal cause of the historic population crash.
- India banned veterinary use of diclofenac, while additional bans introduced in 2023 targeted aceclofenac and ketoprofen to reduce pharmaceutical threats to vultures.
- An official survey reported in 2025 found vultures nesting at only 50% of their historic nesting sites, highlighting the continuing fragility of their recovery.

KEY TAKEAWAY

Infrastructure development and biodiversity conservation must advance together by integrating wildlife behaviour, ecological vulnerability and preventive safeguards into power-system planning.